DATA ANALYTICS EXAMINATION

Excel | SQL Server (SSMS) | Power BI

Total Marks	100	**Duration**	2 Hours
Difficulty	Intermediate	**Exam Type**	Combined Practical + Scenario-Based

Candidate Name: _____ Date: _____ Batch: _____

Examination Structure

Section	Subject	Marks	Suggested Time
A	Microsoft Excel	30	35 minutes
B	SQL Server (SSMS)	35	40 minutes
C	Power BI	35	45 minutes
	TOTAL	**100**	**120 minutes**

General Instructions

1. Attempt all questions.

2. Use Excel, SQL Server Management Studio (SSMS), and Power BI Desktop for practical tasks where available.

3. Use the provided business scenarios and datasets. If a dataset is not supplied separately, create the sample tables or data required by the question.

4. For SQL questions, write complete executable queries using SQL Server syntax.

5. For Excel, use appropriate formulas and functions rather than manually calculating results.

6. For Power BI, demonstrate data transformation, data modelling, DAX calculations, and report design as requested.

7. Save your work with your name and section. Submit the workbook, SQL script, and Power BI file if the exam is conducted on a computer.

8. No solutions or hints are included in this question paper.

Excel Dataset — Sample Records

Use the supplied Data_Analytics_Exam_Dataset.xlsx workbook for the full dataset. The following ten records illustrate the data structure.

OrderID	OrderDate	Region	Product	Category	Quantity	UnitPrice	UnitCost
1001	05-Jan-2026	West	Laptop	Electronics	2	55000	46000
1002	07-Jan-2026	North	Chair	Furniture	5	4500	3000
1003	10-Jan-2026	South	Mouse	Accessories	10	700	400
1004	15-Feb-2026	West	Monitor	Electronics	3	12000	9500
1005	18-Feb-2026	East	Desk	Furniture	2	8500	6200
1006	22-Feb-2026	North	Keyboard	Accessories	8	1500	900
1007	03-Mar-2026	South	Laptop	Electronics	1	55000	46000
1008	12-Mar-2026	West	Chair	Furniture	4	4500	3000
1009	19-Mar-2026	East	Monitor	Electronics	2	12000	9500
1010	25-Mar-2026	North	Mouse	Accessories	15	700	400

SECTION A — MICROSOFT EXCEL (30 MARKS)

Business scenario: Retail Sales Analysis. The sales manager wants to analyze revenue, profitability, sales performance, and customer purchasing patterns.

Question 1 — Calculated columns (5 marks)

Using formulas, add the following columns to the dataset:

1. Calculate total sales revenue for each order.
2. Calculate total cost for each order.
3. Calculate profit for each order.
4. Calculate profit margin as a percentage of revenue.
5. Format the calculated columns appropriately.

Question 2 — Conditional and logical functions (5 marks)

The management team wants to classify orders based on profitability.

1. Create a column that displays High Profit when order profit is greater than ■10,000 and Standard Profit otherwise.
2. Identify orders with revenue greater than ■20,000.
3. Use conditional formatting to highlight orders with negative profit, if any.
4. Apply suitable number formats to revenue and profit.

Question 3 — Conditional aggregation (5 marks)

Prepare a regional sales summary showing:

1. Total revenue for each region.
2. Total profit for each region.
3. Total quantity sold for each region.
4. Number of orders from each region.

Use appropriate Excel functions to calculate the results.

Question 4 — Lookup functions (5 marks)

Create a separate product master table containing Product, Category, UnitPrice, and UnitCost.

1. Retrieve the category for each order using a lookup function.
2. Retrieve the product cost from the master table.
3. Identify any product names that do not match the master table.
4. Ensure the formulas remain usable when new order rows are added.

Question 5 — PivotTable and PivotChart (5 marks)

Create a PivotTable that summarizes:

• Revenue by region and category.

• Profit by product.

• Monthly revenue by order date.

Create an appropriate PivotChart to present one of these summaries to management.

Question 6 — Business insights (5 marks)

Using your completed Excel analysis, prepare a short management summary identifying:

1. The highest-revenue region.
2. The most profitable product.
3. The month with the highest sales.
4. One business recommendation based on the results.

Support your conclusions with calculated values from your workbook.

SECTION B — SQL SERVER (SSMS) (35 MARKS)

Business scenario: E-commerce Order Management. Management needs reports to monitor sales, customer activity, product performance, and regional revenue.

Question 7 — Basic filtering and sorting (5 marks)

Write SQL queries to:

1. Retrieve all orders placed during January 2026.

2. Display all products in the Electronics category with a unit price greater than ■10,000.

3. Display the five most expensive products in descending order of price.

4. Retrieve orders whose status is either Pending or Shipped.

Question 8 — Joins and customer analysis (5 marks)

Write a query that displays Order ID, order date, customer name, customer region, product name, quantity ordered, and selling price per unit. Combine the required tables and return the complete order-level product details.

Question 9 — Aggregation and grouping (5 marks)

Write queries to calculate:

1. Total revenue by product category.

2. Total quantity sold by product.

3. Total number of orders placed by each customer.

4. Average order value for each region.

Use appropriate aggregation and grouping techniques.

Question 10 — HAVING and business performance (5 marks)

The sales manager wants to identify high-performing products.

Write a query that returns products with total sales revenue greater than ■50,000. Display the product name, total quantity sold, total revenue, and total profit.

Calculate revenue using OrderDetails.Quantity × OrderDetails.UnitPrice and profit using (selling price – product unit cost) × quantity.

Question 11 — Common Table Expression (CTE) (5 marks)

Using a CTE, calculate total revenue generated by each customer.

Return the customer ID, customer name, total revenue, and customer ranking from highest to lowest revenue.

Question 12 — Date-based sales analysis (5 marks)

Write a query that summarizes sales performance by month for the year 2026.

Include year and month, total orders, total quantity sold, total revenue, and total profit.

Ensure that the results are grouped chronologically.

Question 13 — Data quality and integrity (5 marks)

Write SQL queries to identify:

1. Customers who have never placed an order.

2. Products that have never been ordered.

3. Orders that contain no matching order-detail records.

4. Duplicate customer records based on customer name and city.

SECTION C — MICROSOFT POWER BI (35 MARKS)

Business scenario: Retail Performance Dashboard. Build a report for leadership covering revenue, profit, sales trends, product performance, and regional results.

Question 14 — Data import and transformation (5 marks)

Using Power BI Desktop and Power Query:

1. Import the sales data.
2. Set appropriate data types for dates, quantities, prices, and costs.
3. Identify and handle blank values and duplicate records where appropriate.
4. Rename columns to meaningful business-friendly names.
5. Apply the transformations and load the data into the model.

Question 15 — Data modelling (5 marks)

Create a suitable data model for sales analysis.

1. Establish the necessary relationships between sales, products, and customers.
2. Create or configure a date table for time-based analysis.
3. Set appropriate relationship cardinality and filter direction.
4. Identify the appropriate table structure for product, customer, date, and sales data.
5. Verify that the model supports accurate aggregation without unintended duplication.

Question 16 — DAX measures (5 marks)

Create DAX measures to calculate Total Revenue, Total Cost, Total Profit, Profit Margin Percentage, and Total Orders.

Use measures that respond correctly to report filter selections.

Question 17 — KPI cards and visualizations (5 marks)

Create a report page containing Total Revenue, Total Profit, Total Orders, and Profit Margin KPI cards, plus a Revenue by Region chart and a Revenue by Category chart.

Choose suitable visualization types and format the report for business use.

Question 18 — Time intelligence (5 marks)

Create a monthly sales trend report.

1. Display revenue by month.
2. Calculate month-over-month revenue change.
3. Calculate month-over-month percentage change.
4. Compare the current month's revenue with the previous month.
5. Ensure that the measures work correctly with the date context.

Question 19 — Interactive reporting (5 marks)

Improve the dashboard by adding a Region slicer, Category slicer, and Date slicer; appropriate visual interactions; and a drill-down or drill-through feature for investigating product or regional performance.

Question 20 — Management insights and presentation (5 marks)

Present your completed report to the sales manager. Demonstrate:

1. Which region generates the highest revenue.
2. Which product category generates the highest profit.
3. Whether revenue is increasing or decreasing over time.
4. Which region or category may require management attention.
5. Two actionable recommendations supported by the report data.

Final Submission Checklist

• Excel workbook with formulas, summary, PivotTable and PivotChart.

• SQL script containing queries for Questions 7–13.

• Power BI PBIX file containing the data model, DAX measures and report pages.

• Management insights supported by calculated results.

Examiner's note: This is a question paper only. No answers, SQL solutions, DAX formulas, or hints are included. For the formal computer-based exam, provide the full transaction dataset so students can demonstrate analysis beyond the sample records.